MORADABAD INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE & ENGINEERING
CLASS TEST-1 (4th SEMESTER)
SESSION 2025-26
Set-1

SUBJECT NAME: Theory of Automata & Formal Languages
SUBJECT CODE: BCS402
BRANCH: CSE

MM: 20
DURATINON: 1 HR
SECTION: A, B, C

Q.NO.	1	2	3	4	5	6
CO	CO1	CO2	CO1	CO1	CO1	CO1
BLOOM'S LEVEL	K6	K6	K2	K3	K3	K3

Section A (8 Marks) [3×1=3]

1. **Design the minimal DFA for the following languages:**
   (i) Set of all strings over {a, b} which start and end with different symbols.
   (ii) Set of all strings that contain even no. of a's and even no. of b's over {a, b}
   (iii) Set of all binary strings that are divisible by 3.

[3×1=3]

2. **Design the Regular expression for the following regular languages.**
   (i) Set of binary strings which should have at least one 0 and at least one 1.
   (ii) Set of all strings containing at most 2a's over {a, b}.
   (iii) Set of Strings that accepts all strings having even number of a's over {a, b}.

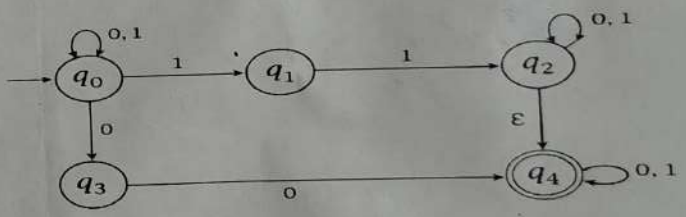
[2]

3. Define DFA and Regular Expression.

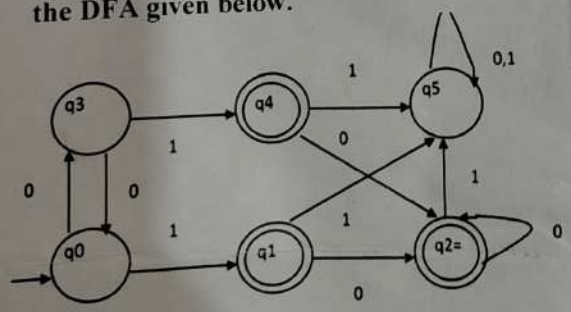
[3x4=12]

Section B (12 Marks)

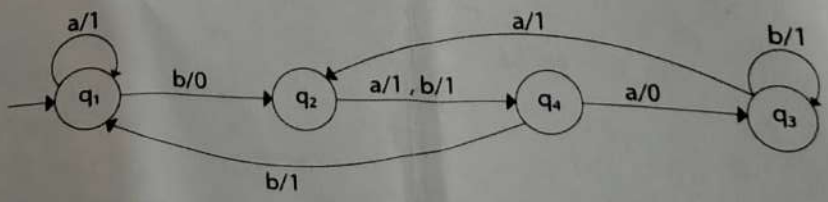
4. **Use Subset Construction Process for obtaining a DFA corresponding to the ϵ - NFA given on the right.**



5. **Construct the minimum state DFA for the DFA given below.**



6. **Convert the below given Mealy machine into equivalent Moore Machine.**



MORADABAD INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE & ENGINEERING
CLASS TEST-2 (4th SEMESTER)
SESSION-2025-26
Set-1

Sub: TOAFL
Sub Code: BCS402
BRANCH: CSE

MM: 20
Duration: 1 hr.
Sec: A, B, C

Q.NO.	1	2	3	4	5	6
CO	CO3	CO3	CO5	CO4	CO4	CO5
BLOOM'S LEVEL	K6	K3	K4	K3	K6	K6

CO3 Understand regular vs. nonregular languages, chomsky hierarchy, design of context free grammars

CO4 Analyses and design pushdown automata

CO5 Understand languages which are not context free and analyses, design simple Turing machines

Section A (08 Marks)

1. Construct a Context Free Grammar (CFG) corresponding to the regular expression $(0 + 1)^*$. [2]

2. Convert the following CFG into Chomsky Normal Form (CNF): [3]
$S \rightarrow aSB \mid \varepsilon$
$B \rightarrow b$.

3. Define Post's Correspondence Problem (PCP). Find any PCP solutions of the lists $x=(b, bab^3, ba)$ and $y=(b^3, ba, a)$. [3]

Section B (12 Marks)

4. Convert CFG given by following productions to equivalent PDA: [4]
$S \rightarrow aAA$, $A \rightarrow aS \mid bS \mid a$
Also, check whether the string 'abaaaa' is in language of PDA or not.

5. Design a PDA for the language: [4]
$L = \{wcw^R \mid w \in \{a, b\}^*\}$.

6. Design a Turing Machine to accept the language: [4]
$L = \{a^n b^n c^n \mid n \geq 1\}$